Prior Attempts

[Komodo Dragons - Nashville Zoo Blog]
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Gough Plastics Crocodile Transport Module

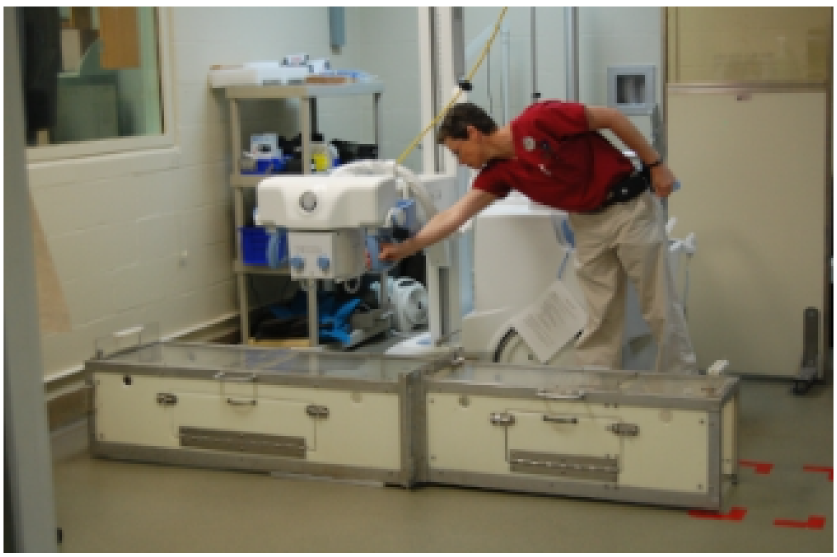
In this video, a crocodile transport module was designed by the Dept. of Environment and Heritage Protection. Their primary purpose was to catch crocodiles in a way that was safer for rangers and less stressful for the crocodiles. It includes a rounded base to hold the crocodile in place and a sprinkler system to keep the crocodile wet.

The box pictured in the photo is used for trapping Komodo dragons on Komodo Island. Mr. Nick Hanna from the Nashville Zoo is pictured carrying the mobile trap during the Komodo Survival Program. This trap appears to be segmented in the modular fashion requested by the zookeepers as well as having the door style requested for safety. This solution is entirely metal which makes it unsuitable for x-ray scans. Additionally, there is no access to the dragon other than the ends which makes taking an ultrasound complicated or even impossible.

The current problem for the zoo is the safety of both the zoo keepers and the Komodo dragons. The solution needs to be capable of holding multiple sizes of dragons and allows for blood to be drawn, x-rays to be taken, and ultrasounds for the females.
The Nashville Zoo currently uses plastic culvert material/tubes to transport the dragons from their enclosure to the vet clinic. The problems that they face with this include not securely holding the dragons and not having handles on the tubes.
When visiting the Nashville Zoo and talking with Nick Hanna we found out more about the problem. They currently use thick corrugated plastic tubes to move the Komodo dragons.

The shortcomings that they face include the tubes being difficult to carry, and they being unable to see the orientation of the dragon's heads while they are in the box. This creates a dangerous situation for the zookeepers when they have to reach in to work with the dragons. They also use modified riot shields and Y-sticks to get the dragons in the tubes.

Disney World's Animal Kingdom started working with Komodo dragons in 1999. They were able to train the dragons for certain tasks when they were in enclosures, as well as some behaviors for the vet to take care of the dragons. They were able to train them to step on a scale, get their nails trimmed, and go into the crates. The keepers at Disney made a crate to transport the dragons and perform medical procedures on them. They required a way to work with the animals that didn't require anesthesia. The crate (shown below) has a clear plastic top, front, and back so that they can examine the animals closely. They also have manual opening doors on the sides to perform ultrasounds and hands-on examinations of the dragons. These crates are also used for drawing blood, radiographs, colloquial swabs, and hind foot nail trims.



The picture above shows the boxes that Disney created for this purpose. The overall side-door design of this box would largely fit the constraints provided for the Nashville Zoo project to perform ultrasounds on female Komodo dragons. However, the metal components used for the side-door mechanism present challenges for the necessary X-ray scans. The metal fasteners, locks, and hinges used to create the side door would deflect X-rays, creating undesirable white spots on the output image (Helmenstine). These boxes allow for different sizes of dragons to use the same boxes. The clear portions allow the keepers to closely inspect the dragons to care for them. As seen in the picture below.

Nashville Zoo's Prior Attempts:

The zoo's current solution is a black plastic tube with lids for both sides (as pictured above). The zookeepers' current carrying system for the Komodo dragons is unsafe and provides low visibility of the dragons inside. Dragons are lured into the tube with food, often frozen rodents, before being enclosed inside with lids. Then, zookeepers wait for the dragon to calm down, carry them into the med bay, and let them out by taking off the lid. The current problem for the zoo is the safety of both the zoo keepers and the Komodo dragons. The solution needs to be capable of holding multiple sizes of dragons and allow for blood to be drawn, x-rays to be taken, and ultrasounds for the females.

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Anne Marie Helmenstine, Ph.D. “Can You X-Ray Metal?” *ThoughtCo*, ThoughtCo, 13 Jan. 2020, www.thoughtco.com/x-rays-and-metal-interference-608418.